Recursive Estimation

Fabio Bühler, fabuehle@ethz.ch

Version: 7. August 2024

1 Probability Theory

1.1 Random Variables

1.1.1 Discrete Random Variables

DRV – Discrete Random Variable defined by $\mathcal{X}, p_x : \mathcal{X} \to \mathbb{R}$

- $\mathcal{X} \subset \mathbb{Z}$ of all possible outcomes
- $p_x(\bar{x}) \geq 0$ and $\sum_{\bar{x}\in\mathcal{X}} p_x(\bar{x}) = 1$ or $\sum_{\bar{x}\in\mathcal{X}} p_{x|y}(\bar{x}|\bar{y}) = 1$

Margin. / Sum Rule	$p(\bar{x}) = \sum_{\bar{y}\in\mathcal{Y}} p_{xy}(\bar{x},\bar{y})$
Cond. / Product Rule	$p_{x\,|\,y}(\bar{x}|\bar{y}) := \frac{p_{xy}(\bar{x},\bar{y})}{p_y(\bar{y})}$
Total Prob. Theorem	$p_x(\bar{x}) := \sum_{\bar{y}\in\mathcal{Y}} p_{x\,|\,y}(\bar{x}|\bar{y})p_y(\bar{y})$

1.1.2 Continuous Random Variables

CRV – Continuous Random Variable: DRV $\sum \rightsquigarrow$ CRV $\int$

- **Set** $\mathcal{X} \subseteq \mathbb{R}$ section of real line
- **PDF** p_x satisfies $p_x(\bar{x}) \geq 0$ and $\int_{\mathcal{X}} p_x(\bar{x})d\bar{x} = 1$

Caution: $Pr(x = \bar{x}) = 0, \quad Pr(x \in (a,b)) = \int_a^b p(\bar{x})d\bar{x}$

1.1.3 Conditional PDF

Conditional PDF

Margin. / Sum Rule	$p(\bar{x}|\bar{z}) = \sum_{\bar{y}\in\mathcal{Y}} p_{xy\,|\,z}(\bar{x},\bar{y}|\bar{z})$
Cond. / Product Rule	$p_{x\,|\,yz}(\bar{x}|\bar{y},\bar{z}) := \frac{p_{xy\,|\,z}(\bar{x},\bar{y}|\bar{z})}{p_{y\,|\,z}(\bar{y}|\bar{z})}$

Independence

$$p(x|y) = p(x) \quad \Leftrightarrow \quad p(y|x) = p(y) \quad \Leftrightarrow \quad p(x,y) = p(x)p(y)$$

$p(x,y,z) = p(x,y|z)p(z) \to$ x,y indp. $p(x,y|z) = p(z|x,y)$

Conditional Independence

The knowledge of z makes x and y independent:

$$p(x|y,z) = p(x|z) \quad \Leftrightarrow \quad p(x,y|z) = p(x|z)p(y|z)$$

Caution!!! in general we still have: $p(x,y) \neq p(x)p(y)$

Caution!!! Independence $\not\Rightarrow$ Conditional Independence

1.2 Expectation and Variance

1.2.1 Expectation

Definition: *Integral for CRV*

$$\mathbb{E}_x[x] = \sum_{\bar{x}\in\mathcal{X}} \bar{x}p_x(\bar{x})$$

- **Linearity**: $\mathbb{E}_{xy}[a + bx + cy] = a + b\,\mathbb{E}_x[x] + c\,\mathbb{E}_y[y]$
- **Multi Variable**: $\mathbb{E}_{xy}[g(x,y)] = \sum_{\bar{y}}\sum_{\bar{x}} g(\bar{x},\bar{y})p_{xy}(\bar{x},\bar{y})$
- **Intependence of** x, $y \Rightarrow \mathbb{E}_{xy}[xy] = \mathbb{E}_x[x]\,\mathbb{E}_y[y]$

Law of Unconcious Statistician for $y = g(x)$

$$\mathbb{E}_y[y] = \sum_{\bar{y}\in\mathcal{Y}} \bar{y}p_y(\bar{y}) = \sum_{\bar{x}\in\mathcal{X}} g(\bar{x})p_x(\bar{x})$$

1.2.2 Variance (generally a matrix)

$$\underset{x}{\mathrm{Var}}[x] = \underset{x}{\mathbb{E}}\left[(x - \mathbb{E}_x[x])(x - \mathbb{E}_x[x])^\top\right] \overset{1D}{=} \underset{x}{\mathbb{E}}\left[x^2\right] - \underset{x}{\mathbb{E}}[x]^2$$

Linearity

$$\mathrm{Var}_x[a + bX + cY] = b^2\,\mathrm{Var}_X[x] + c^2\,\mathrm{Var}_Y[y] + 2bc\mathrm{Cov}[X,Y]$$

Covariance: if independent $\mathrm{Cov}(X,Y) = 0$

$$\mathrm{Cov}[X,Y] = \mathbb{E}[(X - \mathbb{E}[X])(Y - \mathbb{E}[Y])]$$

1.3 Distribution Sampling

CDF – Cumulative Distribution Function

$$\widehat{F}_x(\bar{x}) := \sum_{i=-\infty}^{\bar{x}} \widehat{p}_x(i) = Pr(x \leq \bar{x}) = \int_{-\infty}^{\hat{x}} p_x(\hat{x})d\hat{x}$$

A1 – 1 DRV $\bar{u}$ sample of $u \sim \mathcal{U}$, solve for $\bar{x}$ such that

$$\widehat{F}_x(\bar{x} - 1) < \bar{u} \leq \widehat{F}_x(\bar{x})$$

A2 – Multiple DRV or CRV

Works for finite & infinite # of elements

- Decompose $\widehat{p}(\bar{x},\bar{y}) = \widehat{p}_{x\,|\,y}(\bar{x}|\bar{y})\widehat{p}_y(\bar{y})$
- Apply **A1** to get sample $\bar{y}$ via $\widehat{p}_y(\bar{y})$
- With $\bar{y}$ fixed, apply **A1** to get sample $\bar{x}$ via $\widehat{p}_{x\,|\,y}(\bar{x}|\bar{y})$

A3 – 1 CRV $\widehat{p}_x$: pw-conti., bounded. $u \sim \mathcal{U}(0,1)$ and solve:

$$\bar{u} = F_x(\bar{x}) \to \bar{x} = F_x^{-1}(\bar{u})$$

1.4 Change of Variables

DRV: $x = g(y)$

$$p_x(\bar{x}) = \sum_{\bar{y}\in\mathcal{Y}: g(\bar{y})=\bar{x}} p_y(\bar{y})$$

CRV general case $y = g(x), \to x = g^{-1}(y)$:

$$p_y(\bar{y}) = \frac{p_x(\bar{x})}{\left|\frac{dg}{dx}(\bar{x})\right|}, \quad \bar{x} = g^{-1}(y)$$

CRV – Multivar. & Indep. Case: $z = x + y$

$$p_z(\bar{z}) = \sum_{\bar{x}\in\mathcal{X}} p_x(\bar{x})p_y(\bar{z} - \bar{x})$$

CRV – Multivar. & Cond. Case: $z = g(w,x) \rightsquigarrow w = h(z,x)$

Assumptions:
- $g(w,x), p_w$ cont. diff'bar
- $g(w,x)$ strictly monotonic
- For scalars $det(x)$ is just x

$$p_{z\,|\,x}(\bar{z}|\bar{x}) = p_{w\,|\,x}(h(\bar{z},\bar{x})|\bar{x}) \cdot \left|\det\left(\frac{\partial}{\partial w}g(h(\bar{z},\bar{x}),\bar{x})\right)\right|^{-1}$$

CRV – Multivar. & Cond. Case: $z = g(w) \rightsquigarrow w = h(z)$

$$p_z(\bar{z}) = p_w(h(\bar{z})) \cdot \left|\det\left(\frac{\partial}{\partial w}g(w)\Big|_{w=h(\bar{z})}\right)\right|^{-1}$$

1.5 Miscellaneous

Indep. multiplication: $z = xy \to p(z) = \sum_Y p_x(\frac{z}{y})p_y(y)$

Uniform Distribution

$$p(x) = \begin{cases} \frac{1}{b-a} & a \leq x \leq b \\ 0 & \text{else} \end{cases} \qquad \mathbb{E}\{x\} = \frac{a+b}{2} \qquad \mathrm{Var}\{x\} = \frac{(b-a)^2}{12}$$

Triangular Distribution

$$p(x) = \begin{cases} \frac{2(x-a)}{(b-a)(c-a)} & a \leq x \leq c \\ \frac{2(b-x)}{(b-a)(b-c)} & c \leq x \leq b \\ 0 & \text{else} \end{cases}$$

$$F(x) = \begin{cases} 0 & x < a \\ \frac{(x-a)^2}{(b-a)(c-a)} & a \leq x \leq c \\ 1 - \frac{(b-x)^2}{(b-a)(b-c)} & c \leq x \leq b \\ 1 & x > b \end{cases}$$

$$\mathbb{E}\{x\} = \frac{a+b+c}{3}, \quad \mathrm{Var}\{x\} = \frac{a^2+b^2+c^2-ab-ac-bc}{18}$$

Gaussian Distribution $\mathbf{x} = (x_1, \ldots, x_D) \sim \mathcal{N}(\boldsymbol{\mu}, \boldsymbol{\Sigma})$

$$p_{\mathbf{x}}(\bar{\mathbf{x}}) = \frac{1}{(2\pi)^{D/2}\det(\boldsymbol{\Sigma})^{1/2}} \exp\left(-\frac{(\bar{\mathbf{x}} - \boldsymbol{\mu})^T\boldsymbol{\Sigma}^{-1}(\bar{\mathbf{x}} - \boldsymbol{\mu})}{2}\right)$$

$\rightsquigarrow$ fully characterized by mean $\boldsymbol{\mu} \in \mathbb{R}^D$ and Variance $\boldsymbol{\Sigma} \in \mathbb{R}^{D\times D}$

Affine transformation of Gaussian: $\boldsymbol{x} \sim \mathcal{N}(\boldsymbol{\mu}, \boldsymbol{\Sigma})$

$$\boldsymbol{y} = M\boldsymbol{x} + \boldsymbol{b} \quad \to \quad \boldsymbol{y} \sim \mathcal{N}(M\boldsymbol{\mu} + \boldsymbol{b}, M\boldsymbol{\Sigma}M^T)$$

Linear combination of jointly GRV: $\boldsymbol{x}, \boldsymbol{y}$

$$\boldsymbol{z} = M_x\boldsymbol{x} + M_y\boldsymbol{y} \quad \Rightarrow \quad \boldsymbol{z} \sim \mathcal{N}(\boldsymbol{\mu}_z, \boldsymbol{\Sigma}_z)$$

$$\boldsymbol{\mu}_z = M_x\boldsymbol{\mu}_x + M_y\boldsymbol{\mu}_y$$

$$\boldsymbol{\Sigma}_z = M_x\boldsymbol{\Sigma}_x M_x^T + M_x\boldsymbol{\Sigma}_{xy}M_y + M_y\boldsymbol{\Sigma}_{xy}^T M_x + M_y\boldsymbol{\Sigma}_y M_y^T$$

(Red symbols are zero if $\boldsymbol{x}, \boldsymbol{y}$ are independent)

Diagonal Variance $\rightsquigarrow$ y_i mut.indep. iff Σ diagonal

$$p(y) = \prod_{i=1}^{D} \frac{1}{\sqrt{2\pi\sigma_i^2}} \exp\left(-\frac{(y_i - \mu_i)^2}{2\sigma_i^2}\right)$$

Scalar Gaussian Distribution $D = 1$, $\sigma^2 = \det(\Sigma) \to \det(\Sigma)^{1/2} = \sigma$

$$p(x) = \frac{1}{\sqrt{2\pi\sigma^2}} \exp\left(-\frac{(x-\mu)^2}{2\sigma^2}\right) \qquad \mathbb{E}\{x\} = \mu \qquad \mathrm{Var}\{x\} = \sigma^2$$

2 Bayesian Tracking

2.1 Bayes Theorem

$$p(x|y) = \frac{p(y|x)p(x)}{p(y)}$$

x=system state, $p(x)$=prior belief of state, z=observations related to state
$p(z|x)$=observation likelihood, $p(x|z)$=posterior belief of state
$p(z)$=normalization constant $= \sum_x p(z|x)p(x)$

For multiple observations:

$$\underbrace{p(x|z_1,\ldots,z_N)}_{\text{posterior}} = \frac{\overbrace{p(x)}^{\text{prior}}\ \overbrace{\prod_i p(z_i|x)}^{\text{observation likelihood}}}{\underbrace{\sum_{x\in\mathcal{X}} p(x)\prod_i p(z_i|x)}_{\text{normalization}}}$$

2.2 Bayesian Tracking

Model

Objective: Calculate pdf of state $p(x(k)|z(1:k))$

$$x(k) = q_{k-1}(x(k-1), v(k-1)) \in \mathcal{X}$$

$$z(k) = h_k(x(k), w(k))$$

Prior Update - State Prediction

$$p(x(k)|z(1:k-1)) = \sum_{x(k-1)} \underbrace{p(x(k)|x(k-1))}_{\text{process model}}\ \underbrace{p(x(k-1)|z(1:k-1))}_{\text{previous iteration}}$$

Measurement Update

$$p(\bar{x}(k)|\bar{z}(1:k)) = \frac{\overbrace{p(\bar{z}(k)|\bar{x}(k))}^{\text{measurement model}}\ \overbrace{p(\bar{x}(k)|\bar{z}(1:k-1))}^{\text{prior}}}{\underbrace{\sum_{i\in\mathcal{X}} p(\bar{z}(k)|i)p(i|\bar{z}(1:k-1))}_{\text{normalization}}}$$

Computer Implementation

1. **Enumerate** state space $\mathcal{X} = \{0, 1, \ldots, N-1\}$
2. **Posterior** define $\mathbf{a}_{k|k}^i := p_{x(k)|z(1:k)}(i|\bar{z}(1:k))$
3. **Prior** define $\mathbf{a}_{k|k-1}^i := p_{x(k)|z(1:k-1)}(i|\bar{z}(1:k-1))$

Initialization $\mathbf{a}_{0|0}^i = p_{x(0)}(i) \quad i = 0, \ldots, N-1$

$$\left.\begin{aligned} \mathbf{a}_{k|k-1}^i &= \textstyle\sum_{j=0}^{N-1} p_{x(k)|x(k-1)}(i|j)\mathbf{a}_{k-1|k-1}^j \\ \mathbf{a}_{k|k}^i &= \frac{p_{z(k)|x(k)}(\bar{z}(k)|i)\mathbf{a}_{k|k-1}^i}{\sum_{j=0}^{N-1} p_{z(k)|x(k)}(\bar{z}(k)|j)\mathbf{a}_{k|k-1}^j} \end{aligned}\right\} \text{Iterate k} > 0$$

3 Extracting Estimates

3.1 Likelihood Estimators

3.1.1 ML-Maximum Likelihood

Choose x to make $\bar{z}$ most likely:

$$\left.\begin{aligned} x &\in \mathcal{X} \quad \text{unknown} \\ p_x(\bar{x}) &\quad \text{unknown} \end{aligned}\right\} \quad \widehat{x}^{ML} := \underset{\bar{x}\in\mathcal{X}}{\mathrm{argmax}}\left[p_{z\,|\,x}(\bar{z}|\bar{x})\right]$$

$p_{z\,|\,x}(\bar{z}|\bar{x})$ is var. – technically not a PDF

3.1.2 MAP-Maximum A Posteriori

ML given observation & prior belief about x:

$$\left.\begin{aligned} x &\in \mathcal{X} \quad \text{unknown} \\ p_x(\bar{x}) &\quad \textbf{known} \end{aligned}\right\} \quad \widehat{x}^{MAP} := \underset{\bar{x}\in\mathcal{X}}{\mathrm{argmax}}\left[p_{z\,|\,x}(\bar{z}|\bar{x})p_x(\bar{x})\right]$$

$\hat{x}^{MAP} = \hat{x}^{ML}$ for constant $p_x(\bar{x})$

3.2 Least Squares Estimators $z, w \in \mathbb{R}^m, x \in \mathbb{R}^n, H \in \mathbb{R}^{m\times n}$

$$z = Hx + w$$

3.2.1 Linear Least Squares

$$\widehat{x}^{LS} := \underset{x}{\mathrm{argmin}}\left[(\bar{z} - Hx)^\top(\bar{z} - Hx)\right] = (H^\top H)^{-1}H^\top\bar{z}$$

$\hat{x}^{LS} = \hat{x}^{ML}$, when $w \sim \mathcal{N}(0, \sigma^2\mathbb{I})$

3.2.2 Weighted Least Squares

$$\widehat{x}^{WLS} := \mathrm{argmin}_x\left[(\bar{z} - H\widehat{x})^\top\Sigma^{-1}(\bar{z} - H\widehat{x})\right]$$

$$= (H^\top\Sigma^{-1}H)^{-1}H^\top\Sigma^{-1}\bar{z}$$

$\hat{x}^{WLS} = \hat{x}^{ML}$, when $w \sim \mathcal{N}(0, \Sigma)$

Incorporate Prior Knowledge on x:

$r \sim \mathcal{N}(0, P_x) \Rightarrow x := \widehat{x} + r, \mathbb{E}\{x\} = \widehat{x}_0, \mathrm{Var}\{x\} = P_x$

Introduce Extended System:

$$\bar{\tilde{z}} = \begin{bmatrix} \hat{x}_0 \\ z(1:k) \end{bmatrix}, \tilde{H} = \begin{bmatrix} \mathbb{I} \\ H(1:k) \end{bmatrix} \tilde{w} = \begin{bmatrix} -r \\ w(1:k) \end{bmatrix}, \tilde{\Sigma} = \begin{bmatrix} P_x & 0 \\ 0 & R(1:k) \end{bmatrix}$$

3.2.3 Recursive Least Squares

Interpretation Update $\widehat{x}$ recursively when new $\bar{z}(k)$ arrives
Estimation Error: $e(k) = (\mathbb{I} - KH)\,e(k-1) - Kw(k)$
Objective Minimize MSE with $P(k) = \mathrm{Var}\{e(k)\}$

$$J(k) := \mathbb{E}\left\{e^\top(k)e(k)\right\} = \mathbb{E}\{\mathrm{trace}(P(k))\}$$

We Have: $\hat{x}_0 = \mathbb{E}[x], P_x = \mathrm{Var}[x], \mathbb{E}[w(k)] = 0, R(k) = \mathrm{Var}[w(k)]$

Initialization: $\widehat{x}(0) = \widehat{x}_0, \quad P(0) = P_x = \mathrm{Var}\{x\}$

Recursion: Observe $\bar{z}(k)$, then update:

$$K(k) = P(k-1)H(k)^\top\left(H(k)P(k-1)H(k)^\top + R(k)\right)^{-1}$$

$$\widehat{x}(k) = \widehat{x}(k-1) + K(k)\left(\bar{z}(k) - H(k)\widehat{x}(k-1)\right)$$

$$P(k) = (\mathbb{I} - K(k)H(k))P(k-1)(\mathbb{I} - K(k)H(k))^\top + K(k)R(k)K(k)^\top$$

4 Kalman Filter

4.1 Kalman Filter

Model (Linear Time-Invariant)

$$x(k) = A(k-1)x(k-1) + u(k-1) + v(k-1)$$

$$z(k) = H(k)x(k) + w(k)$$

$x(0), v(\cdot), w(\cdot)$ ind.	$x(0) \sim \mathcal{N}(x_0, P_0)$
$v \sim \mathcal{N}(0, Q(k))$	$w \sim \mathcal{N}(0, R(k))$

Auxilliary Variables

Problem Formulation want $p_{x(k)|z(1:k)} \rightsquigarrow$ bayesian tracking provides this
BUT we have **CRVs** and don't want integrals $\Rightarrow$ exploit linearity
Auxilliary Variables:

Init: $x_m(0) = x(0)$

S1: $x_p(k) = A(k-1)x_m(k-1) + u(k-1) + v(k-1)$

S2: $z_m(k) = H(k)x_p(k) + w(k)$

$$p_{x_m(k)}(\xi) := p_{x_p(k)|z_m(k)}(\xi|\bar{z}(k)) \quad \forall\xi$$

Equivalency:

$$p_{x_p(k)}(\xi) = p_{x(k)|z(1:k-1)}(\xi|\bar{z}(1:k-1))$$

$$p_{x_m(k)}(\xi) = p_{x(k)|z(1:k)}(\xi|\bar{z}(1:k))$$

GRV x_p and x_m are GRVs for $k = 1, 2, \ldots$

$$\widehat{x}_p(k) := \mathbb{E}\left[x_p(k)\right] \qquad P_p(k) := \mathrm{Var}\left[x_p(k)\right]$$

$$\widehat{x}_m(k) := \mathbb{E}\left[x_m(k)\right] \qquad P_m(k) := \mathrm{Var}\left[x_m(k)\right]$$

Recursive Update Equations

Initialization $\widehat{x}_m(0) = x_0, \quad P_m(0) = P_0$
S1 – Prior Update / Prediction Step

$$\widehat{x}_p(k) = A(k-1)\widehat{x}_m(k-1) + u(k-1)$$

$$P_p(k) = A(k-1)P_m(k-1)A^\top(k-1) + Q(k-1)$$

S2 – A Posteriori Update / Measurement Update

$$P_m(k) = \left(P_p^{-1}(k) + H^\top(k)R^{-1}(k)H(k)\right)^{-1}$$

$$\widehat{x}_m(k) = \widehat{x}_p(k) + \underbrace{P_m(k)H^\top(k)R^{-1}(k)}_{\text{correctionfactor}}\underbrace{\left(\bar{z}(k) - H(k)\widehat{x}_p(k)\right)}_{\text{predictionerror}}$$

Alternative Equations – all matrices/vectors at time k

$$K(k) = P_p(k)H^\top(k)\left(H(k)P_p(k)H^\top(k) + R(k)\right)^{-1}$$

S2 – Kalman Filter Gain:

$$\widehat{x}_m = \widehat{x}_p + K\left(\bar{z} - H\widehat{x}_p\right)$$

$$P_m = \underbrace{(\mathbb{I} - KH)\,P_p}_{\text{cheaper}} = \underbrace{(\mathbb{I} - KH)\,P_p\,(\mathbb{I} - KH)^\top + KRK^\top}_{\text{less sensitive to num. errors}}$$

KF Remarks

- The KF is the analytical solution to the Bayesian state estimation problem for linear systems with Gaussian distributions. (Best You can do)
- Otherwise, the KF is no longer optimal. It is the best linear estimator in the MMSE sense but nonlinear might do better.
- $A(k), H(k), Q(k), R(k) \quad \forall k$ as well as $P_0 \rightsquigarrow$ KF matrices $P_p(k), P_m(k), K(k)$ can be computed offline, K no f() of Measurm.
- KF is linear and time-varying:

$$\widehat{x}_m(k) = \tilde{A}(k) \cdot \widehat{x}_m(k-1) + \tilde{B}(k) \cdot u(k-1) + K\bar{z}(k)$$

$$\tilde{A}(k) = (\mathbb{I} - K(k)H(k))A(k-1), \quad \tilde{B}(k) = (\mathbb{I} - K(k)H(k))$$

- Among class of linear, unbiased estimators, the KF minimizes MSE
- Not all system matrices need to be pd. some can also be psd. as long as inversions are possible in the update equations.

4.1.1 Detectability / Stabilizablilty

Detectability Conditions

Pair (A, H) detectable

$\Leftrightarrow$ For deterministic LTI Sys. $\left(x(k)=Ax(k-1), z(k)=Hx(k)\right)$
$\lim_{k\to\infty} z(k) = 0 \Rightarrow \lim_{k\to\infty} x(k) = 0 \quad \forall x_0 \in \mathbb{R}^n$

$\Leftrightarrow$ $\begin{bmatrix} A-\lambda\mathbb{I} \\ H \end{bmatrix}$ is full rank for all $\lambda \in \mathbb{C}$ with $|\lambda| \geq 1$ (PBH-Test)

$\Leftrightarrow$ Eigenvalues of $A - LH$ (or $(\mathbb{I} - LH)A$) can be placed within unit circle by suitable choice of $L \in \mathbb{R}^{n\times m}$

Stabilizability Conditions

Pair (A, B) stabilizable

$\Leftrightarrow$ For deterministic LTI Sys. $\left(x(k)=Ax(k-1)+Bu(k-1)\right)$
$\exists\, u(0:k-1)$s.t. $\lim_{k\to\infty} x(k) = 0 \quad \forall x_0 \in \mathbb{R}^n$

$\Leftrightarrow$ $[\,A-\lambda\mathbb{I} \;\; B\,]$ full rank $\forall\lambda \in \mathbb{C}$ with $|\lambda| \geq 1$ (PBH-Test)

$\Leftrightarrow$ Eigenvalues of $A - BK$ (or $(\mathbb{I} - BK)A$) can be placed within unit circle by suitable choice of $K \in \mathbb{R}^{m\times n}$

Observability / Reachability

if $(A^\top, B^\top)$ detectable

Observability Matrix $\mathcal{O} = \begin{bmatrix} H \\ HA^{n-1} \end{bmatrix}$
implies detectability

Reachability Matrix $\mathcal{R} = [\,A \;\; A^{n-1}B\,]$
implies stabilizablilty

4.1.2 Asymptotic Properties of KF

Model assume **TI System, stationary Distributions** $(A, H, Q, R$ constants$)$
$\rightsquigarrow$ the KF is still time-varying!

Change of Variance

$$P_p(k+1) = \left(AP_p(k)A^\top\right) + Q - \left(AP_p(k)H^\top\right)\left(HP_p(k)H^\top + R\right)^{-1}\left(HP_p(k)A^\top\right)$$

Scalar Case $P_p(k+1) = \frac{a^2 r P_p(k)}{h^2 P_p(k) + r} + q =: f(P_p(k))$

Summary *Variance Converges to unique steady-state sol'n provided either $|a| < 1$ or if $|a| \geq 1 \Rightarrow h \neq 0$ and $q > 0$*

4.1.3 Steady-State KF

DARE – Discrete Algebraic Ricatti Equation (P_∞ is P_p)

$$P_\infty = AP_\infty A^\top + Q - AP_\infty H^\top (HP_\infty H^\top + R)^{-1} HP_\infty A^\top$$

Steady-State KF Gain: $K_\infty = P_\infty H^\top (HP_\infty H^\top + R)^{-1}$

Steady State KF Equations $\widehat{x}(k) := \widehat{x}_m(k)$

$$\widehat{x}(k) = (\mathbb{I} - K_\infty H)A\widehat{x}(k-1) + (\mathbb{I} - K_\infty H)u(k-1) + K_\infty \bar{z}(k)$$

$$\widehat{x}(k) = \widehat{A} \cdot \widehat{x}(k-1) + \widehat{B} \cdot u(k-1) + K_\infty \cdot \bar{z}(k), \quad \widehat{x}(0) = x_0$$

Error dynamics (also holds for non ss-KF) $e(k) = x(k) - x_f(k)$

$$e(k) = (\mathbb{I} - K_\infty H)Ae(k-1) + (\mathbb{I} - K_\infty H)v(k-1) - K_\infty w(k)$$

$$\mathbb{E}[e(k)] = (\mathbb{I} - K_\infty H)A\,\mathbb{E}[e(k-1)]$$

- If $\mathbb{E}[e(0)] = 0$ then $\mathbb{E}[e(k)] = 0 \quad \forall k \rightsquigarrow$ KF is unbiased
- $\mathbb{E}[e(k)] \xrightarrow{k\to\infty} 0 \Leftrightarrow (\mathbb{I} - K_\infty H)A$ stable (all $|\lambda| < 1$)

Theorem for $R > 0, Q \geq 0, G$ s.t. $Q = GG^\top$

$\Leftrightarrow$ (A, H) is detectable and (A, G) is stabilizable

$\Leftrightarrow$ DARE has unique pos. semidef. sol. $P_\infty \geq 0$, resulting $(\mathbb{I} - K_\infty H)A$ is stable and

$$\lim_{k\to\infty} P_p(k) = P_\infty \quad \forall P_p(1) \geq 0 \text{ and } \forall P_m(0) = P_0 \geq 0$$

(A,H) Detectable: can observe all unstable modes

(A,G) Stabilizable: noise excites unstable modes

Note if $Q > 0 \Rightarrow (A, G)$ always stabilizable If $Q = 0$ multiple valid solutions can be possible

4.2 Extended Kalman Filter

$$x(k) = q_{k-1}(x(k-1), v(k-1))$$

$$z(k) = h_k(x(k), w(k)) \qquad x(0), v(\cdot), w(\cdot) \text{ mutually ind.}$$

$$\mathbb{E}[x(0)] = x_0, \mathrm{Var}[x(0)] = P_0 \quad | \quad \mathbb{E}[w(k)] = 0, \mathrm{Var}[w(k)] = R(k)$$

$$\mathbb{E}[v(k-1)] = 0, \mathrm{Var}[v(k-1)] = Q(k-1)$$

EKF Equations

Initialization: $\widehat{x}_m(0) = x_0, \quad P_m(0) = P_0$

S1 – Prior Update / Prediction Step

$\widehat{x}_p(k) = q_{k-1}(\widehat{x}_m(k-1), 0)$

$$P_p(k) = A(k-1)P_m(k-1)A^\top(k-1) + L(k-1)Q(k-1)L^\top(k-1)$$

$$A(k-1) = \frac{\partial q_{k-1}(\widehat{x}_m(k-1), 0)}{\partial x}, \quad L(k-1) = \frac{\partial q_{k-1}(\widehat{x}_m(k-1), 0)}{\partial v}$$

S2 – A Posteriori Update / Measurement Update (at k)

$$K = P_p H^\top \left(HP_pH^\top + MRM^\top\right)^{-1}$$

$$\widehat{x}_m = \widehat{x}_p + K\left(\bar{z} - h_k(\widehat{x}_p(k), 0)\right)$$

$$P_m = (\mathbb{I} - K(k)H(k))\,P_p(k)$$

$$H(k) := \left.\frac{\partial h_k(x, w)}{\partial x}\right|_{\substack{x=\widehat{x}_p(k)\\ w=0}}, \quad M(k) := \left.\frac{\partial h_k(x, w)}{\partial w}\right|_{\substack{x=\widehat{x}_p(k)\\ w=0}}$$

Caution

- Process & measurement noise **assumed** 0-mean
- A, L, H, M lin. about current $\widehat{x}_m \rightsquigarrow$ **cannot** compute offline
- Prior update only accurate if $\mathbb{E}\{\cdot\}$ & $q_{k-1}(\cdot)$ commute
- **EKF Variables DO NOT capture true conditional mean & var.**

4.2.1 Hybrid EKF

System Process – non-lin, CT, **Measurement** – non-lin, DT, $x[k] := x(kT)$

$\dot{x}(k) = q(x(t), v(t), t)\mathbb{E}[x(0)] = x_0, \mathrm{Var}[x(0)] = P_0$

$$\mathbb{E}[v(t)] = 0, \mathrm{Var}[v(t)v(t+\tau)^\top] = Q_c\delta(\tau)$$

$$z[k] = h_k(x[k], w[k]) \quad \mathbb{E}\{w[k]\} = 0, \mathrm{Var}[w[k]] = R$$

CT White Noise $\delta(t)$ = Dirac pulse

$$\mathbb{E}\{v(t)\} = 0 \text{ and } \mathbb{E}\{v(t)v^\top(t+\tau)\} = \mathcal{Q}_c\delta(\tau)$$

True CT White noise does not exist $\rightarrow$ Would require infinite power

Hybrid EKF Equations

Initialization $\widehat{x}_m(0) = x_0, \quad P_m(0) = P_0$

S1 – Prior Update / Prediction Step (Solve ODEs)

Solve, for $(k-1)T \leq t \leq kT$ and $\hat{x}((k-1)T) = \widehat{x}_m[k-1]$

$$\dot{\hat{x}}(t) = q(\hat{x}(t), 0, t) \quad \Rightarrow \hat{x}_p[k] := \hat{x}(kT)$$

Solve, for $(k-1)T \leq t \leq kT$ and $P((k-1)T) = P_m[k-1]$

$$\dot{P}(t) = AP + PA^\top + LQ_cL^\top \quad \Rightarrow P_p[k] := P(kT)$$

$$A(t) := \frac{\partial q}{\partial x}(\hat{x}(t), 0, t), \qquad L(t) := \frac{\partial q}{\partial v}(\hat{x}(t), 0, t)$$

S2 – A Posteriori Update / Measurement Update

Measurement model still DT – S2 identical to standard EKF

Caution

- Process & measurement noise **assumed** 0-mean
- ODEs typically solved numerically: Accuracy $\uparrow$ = Comp. Cost $\uparrow$

5 Particle Filter

5.1 Monte Carlo Sampling

Samples N DRVs y^n of samples of $y \in \mathcal{Y} = \{1, \ldots, \bar{Y}\} \sim p_y$

Helper Variables

$$s_i^n = \delta(i - y^n) = \begin{cases} 1 & \text{if} y^n = i \\ 0 & \text{otherwise} \end{cases} \quad \left| \begin{array}{l} i = 1, \ldots, \bar{Y} \\ n = 1, \ldots, N \end{array} \right.$$

Average – s_i converges to $p_y(i)$ by Law of Large Numbers

$$\frac{1}{N}\sum_{n=1}^{N} s_i^n \xrightarrow{N\to\infty} \mathbb{E}\{s_i^n\} = p_y(i) \qquad p_y(i) \approx \frac{1}{N}\sum_{n=1}^{N} \bar{s}_i^n$$

Approximation

For $y \in \mathcal{Y}$, it holds $\quad p_y(i) \approx \frac{1}{N}\sum_{n=1}^{N}\delta(i - \bar{y}^n), \quad i \in \mathcal{Y}$

For $x = g(y)$, with $x \in \mathcal{X} := g(\mathcal{Y})$, it holds

$$p_x(j) \approx \frac{1}{N}\sum_{n=1}^{N}\delta(j - g(\bar{y}^n)), \quad j \in g(\mathcal{Y})$$

Monte Carlo Sampling CRV

Helper Variables

$$s_a^n = \int_a^{a+\Delta y}\delta(i - y^n)d\xi = \begin{cases} 1 & \text{if} a \leq y^n < a + \Delta y \\ 0 & \text{otherwise} \end{cases}$$

Average – s_a converges to $p_y(\xi)$ by Law of Large Numbers

$$\int_a^{a+\Delta y}\frac{1}{N}\sum_{n=1}^{N}\delta(\xi - y^n)d\xi \xrightarrow{N\to\infty} \int_a^{a+\Delta y} p_y(\xi)d\xi$$

Approximation

$$\text{For } y \in \mathcal{Y}\text{, it holds } p_y(\xi) \approx \frac{1}{N}\sum_{n=1}^{N}\delta(\xi - \bar{y}^n)\forall\xi$$

$$\text{For } x = g(y)\text{, it holds } p_x(\xi) \approx \frac{1}{N}\sum_{n=1}^{N}\delta(\xi - g(\bar{y}^n))\forall\xi$$

5.2 Modeling

System *Non-linear, discrete-time*:

$$x(k) = q_{k-1}(x(k-1), v(k-1))$$

$$z(k) = h_k(x(k), w(k))$$

$x(0), \{v\}, \{w\}$ mutually indep. can be D- or CRVs, known PDFs

Objective

Bayesian State Estimator with Monte Carlo Sampling

Init:	$x_m(0)$	$:= x(0)$
S1:	$x_p(k)$	$:= q_{k-1}(x_m(k-1), v(k-1))$
S2:	$z_m(k)$	$:= h_k(x_p(k), w(k))$
	$p_{x_m(k)}(\xi)$	$:= p_{x_p(k)\mid z_m(k)}(\xi\mid\bar{z}(k)) \quad \forall\xi$

Similar proof for $p_{x_p(k)}(\xi), p_{x_m(k)}(\xi)\forall\xi$ as for KF aux. var.

5.3 Particle Filter - Equations

Initialization Draw N samples $\{\bar{x}_m^n(0)\}$ from $p_{x(0)}$

S1 – Prior Update / Prediction Step

Apply process equation to the particles $\{\bar{x}_m^n(k-1)\}$

$$\bar{x}_p^n(k) := q_{k-1}(\bar{x}_m^n(k-1), \bar{v}^n(k-1)), \quad n = 1, \ldots, N$$

Requires N noise samples from $p_{v(k-1)}$

S2 – A Posteriori Update / Measurement Update

Scale each particle by measurement likelihood:

$$\beta_n = \alpha p_{z(k)\mid x(k)}(\bar{z}(k)\mid\bar{x}_p^n(k)), \qquad n = 1, \ldots, N$$

with α normalization s.t $\sum_{n=1}^{N}\beta_n = 1$

$$\alpha = \left(\sum_{n=1}^{N} p_{z_m(k)\mid x_p(k)}(\bar{z}(k)\mid\bar{x}_p^n(k))\right)^{-1}$$

Resample to get N posterior particles $\{\bar{x}_m^n(k)\}$, w/ eq. weights

Repeat N times:
- Select random number r uniformly on $(0, 1)$
- Pick particle $\bar{n}$ s.t $\sum_{n=1}^{\bar{n}-1}\beta_n < r \leq \sum_{n=1}^{\bar{n}}\beta_n$

New particles all have equal weight

$$p_{x_m(k)}(\xi) \approx \frac{1}{N}\sum_{n=1}^{N}\delta(\xi - \bar{x}_m^n(k)), \quad \forall\xi$$

Intuition

Process Update – Propagate N particles through system dynamics. Provided N large, $\{x_m^n(k-1)\} \approx p_{x_m(k-1)} \rightsquigarrow \{\bar{x}_p^n(k)\} \approx p_{x_p(k)}$

Measurement Update – treat each particle separately. At points of high prior, we have many particles. Posterior has same particles, but scaled by measurement likelihood.

5.4 Roughening

Sample Impoverishment *Lumpiness* – when we resample, only retain subset of particles. For finite N this is problem.

Roughening

Perturb particles **after** resampling

$$\bar{x}_m^n(k) \leftarrow \bar{x}_m^n(k) + \Delta x^n(k)$$

$\Delta x^n(k)$ drawn from 0-mean, finite-var. distribution

Example Method for Δx^n: σ_i stand. dev. of $\Delta x_i^n(k)$

$$\sigma_i = KE_iN^{-\frac{1}{d}}$$

K: tuning parameter, typically $K \ll 1$

d: dimension of state space

E_i: max inter-sample variability $\max_{n_1,n_2}|\bar{x}_{m,i}^{n_1} - \bar{x}_{m,i}^{n_2}|$

$N^{-\frac{1}{d}}$: node spacing on grid

6 Observer Based Control

6.1 LTI Observer

LTI System

$$x(k) = Ax(k-1) + Bu(k-1) + v(k-1)$$

$$z(k) = Hx(k) + w(k)$$

v, w 0-mean noise CRVs

$\mathbb{E}\{\widehat{x}(k)\} \xrightarrow{k\to\infty} \mathbb{E}\{x(k)\}$, $\mathrm{Var}\{\widehat{x}(k)\}$ bounded

Luenberger Observer

Same structure as steady-state KF:

$$\begin{aligned}\widehat{x}(k) &= A\widehat{x}(k-1) + Bu(k-1) + K(\bar{z}(k) - \widehat{z}(k)) \\ &= (\mathbb{I} - K_\infty H)A\widehat{x}(k-1) \\ &\quad + (\mathbb{I} - K_\infty H)Bu(k-1) + K_\infty\bar{z}(k) \\ \widehat{z}(k) &= H(A\widehat{x}(k-1) + Bu(k-1))\end{aligned}$$

Error $e(k) = x(k) - \widehat{x}(k)$

in absence of noise $\bar{z}(k) = z(k)$

$$e(k) = (\mathbb{I} - KH)Ae(k-1)$$

- $(\mathbb{I} - KH)A$ must be stable (all $|\lambda| < 1$) for $e \xrightarrow{k\to\infty} 0$
- Such K exists iff (A, H, A) detectable $\Leftrightarrow (A, H)$ detectable

Observer Design

If (A, H) detectable, construct K

Pole Placement: place λ of e-dynamics of observer modes at desired locations

Steady State KF: Yields optimal K (min steady-state MSE) given statistics of v, w. Also need (A, G) stabilizable, $\mathrm{Var}\{v(k)\} = GG^\top$

Alternative Formulation

$$\widehat{x}(k+1) = A\widehat{x}(k) + Bu(k) + K(\bar{z}(k) - \widehat{z}(k))$$

$$\widehat{z}(k) = H\widehat{x}(k) \rightarrow \text{ delayed measurement}$$

Error Dynamics

$$e(k+1) = (A - KH)e(k)$$

6.2 Static State Feeback Control

System Model

LTI, no noise, deterministic, perf. state info $z(k) = x(k)$:

Feedback Law	CL Dynamics
$u(k) = Fx(k) = Fz(k)$	$x(k) = (A + BF)x(k-1)$

- $(A + BF)$ must be stable (all $|\lambda| < 1$)
- Such F exists $\Leftrightarrow (A, B)$ stabilizable

Controller Design

Pole Placement: place (controllable) poles at des. CL loc.

LQR: find F that min. quad. cost ($\bar{Q} = \bar{Q}^\top \geq 0, \bar{R} = \bar{R}^\top \geq 0$)

$$J_{\mathrm{LQR}} = \sum_{k=0}^{\infty} x^\top(k)\bar{Q}x(k) + u^\top(k)\bar{R}u(k)$$

LQR Design

Assuming (A, B) stabil., (A, G) detect.

$$F = -(B^\top PB + \bar{R})^{-1}B^\top PA$$

DARE

Same as KF but $A \to A^\top, H \to B^\top, Q \to \bar{Q}, R \to \bar{R}$

$$P = A^\top PA + \bar{Q} - A^\top PB(B^\top PB + \bar{R})^{-1}B^\top PA$$

6.3 Seperation Principle

Stable Observer + Controller $\rightsquigarrow u(k) = F\widehat{x}(k)$

Dynamics

$$\begin{bmatrix} x(k) \\ e(k) \end{bmatrix} = \begin{bmatrix} A + BF & -BF \\ 0 & (\mathbb{I} - KH)A \end{bmatrix}\begin{bmatrix} x(k-1) \\ e(k-1) \end{bmatrix}$$

Separation Principle

- $(\mathbb{I} - KH)A$ and $(A + BF)$ stable $\rightsquigarrow$ **Overall system stable**
- Still stable with noise, Generalizable to time-varying case
- Does **NOT** hold for nonlinear systems

Separation Theorem

Optimal strategy for control problem

$$J_{\mathrm{LQG}} = \lim_{N\to\infty}\mathbb{E}\left\{\frac{1}{N}\sum_{k=0}^{N-1}\left[x^\top(k)\bar{Q}x(k) + u^\top(k)\bar{R}u(k)\right]\right\}$$

1. Design steady state KF to provide $\widehat{x}(k)$ **(no dependency on $\bar{Q}, \bar{R}$)**
2. Design optimal state-FB law $u(k) = Fx(k)$ for determ. LQR problem minimizes J_{LQR} w/o noise **(no dependency on noise statistics Q, R)**
3. Put together